

Friday Evidence Report

Source: Batch batch_20260611_235330_465964; query MALDI-TOF mass spectrometry antimicrobial resistance detection

Executive Summary

- Friday screened 25 records, blocked 0, and deep-read 3 safe scholarly PDFs.
- The report uses 75 page-anchored evidence rows from 3 parsed paper references.
- Citation audit status: pass; 2 material gaps recorded.
- Most direct result evidence: The reliability and accuracy of MALDI-TOF Grant Medical College and MS in identification of clinically relevant bacteria and yeasts has been demonstrated Sir Jamshedjee Jeejeebhoy Group by several studies showing that the performance of MALDI-TOF MS is comparable or superior to phenotypic methods currently in use in clinical microbiology laboratories, and of Hospitals, India can be further improved by database updates and analysis software upgrades [P2 p1].

Background

- pneumoniae showed a modification of lipid A resistance in a number of randomly selected clinical isolates of by AraN (m/z 131) attached on the terminal phosphate group Mycobacterium tuberculosis and non-tuberculous mycobacteria of hexa-acetylated lipid A (1,824 m/z), resulting in a modified (NTM) [P1 p3]
- Accepted: 22 September 2020 Published: 11 November 2020 Multi-drug-resistance is a particularly dramatic problem in systemic infections (Palacios-Baena et al., 2017), and infections involving critical districts [e.g., eye and orbit, Citation: where timely administration of an effective therapy is fundamental for sparing organ specific Florio W, Baldeschi L, Rizzato C, functions or patient survival (Tsirouki et al., 2018; Choi et al., 2019)] [P1 p1]
- pneumoniae strains, an 11,109-Da peak was detected in the and appearance of peaks representing its hydrolysis products spectra of 30 out of 34 KPC producers, and it was absent in (Burckhardt and Zimmermann, 2011; Hrabak et al., 2011; all non-KPC-producing isolates (Gaibani et al., 2016) [P1 p2]
- by establishing a method, which relies on the production of a phenol-soluble protein toxin (PSM-mec) by a subset of To reduce time for detection of carbapenem-resistance MRSA strains (Chatterjee et al., 2011), which is detectable in Gram-negative bacteria causing bloodstream infections, a by MALDI-TOF MS as a 2415 ? 2.00 m/z peak and may be MALDI-TOF MS-based assay was established by measuring present in up to half of MRSA isolates (Rhoads et al., 2016; the hydrolysis of imipenem in blood cul... [P1 p2]
- preparation protocols have been proposed to separate bacteria from host cells and proteins through cell lysis and/or differential In an attempt to overcome these difficulties, an indirect centrifugation steps (Christner et al., 2010; Stevenson et

al., method has been established to detect methicillin resistance in 2010; Schubert et al., 2011), and differences in sample processing a particular subset of S [P2 p3]

- Thus, the MALDI ? Principle and Methodology identity of a microorganism can be established down to the genus, The sample for analysis by MALDI MS is prepared by mixing or and in many cases to the species and strain level (Fagerquist et al., coating with solution of an energy-absorbent, organic compound 2010) [P3 p3]

Key Findings

- The reliability and accuracy of MALDI-TOF Grant Medical College and MS in identification of clinically relevant bacteria and yeasts has been demonstrated Sir Jamshedjee Jeejeebhoy Group by several studies showing that the performance of MALDI-TOF MS is comparable or superior to phenotypic methods currently in use in clinical microbiology laboratories, and of Hospitals, India can be further improved by database updates and analysis software upgrades [P2 p1]
- However, the results of several studies suggest that the addition of spectra of successfully identified clinical isolates lower score cut-off values might be applied for identification of into commercial databases (Seng et al., 2009; Fern?ndez-Olmos many bacterial species without compromising accuracy (Richter et al., 2012; Nomura, 2015; Carrasco et al., 2016; Mar?n et al., et al., 2012; Szabados et al., 2012; Barnini et al., 2015), and 2017) [P2 p2]
- Microbial identification in polymicrobial samples by MALDI- significantly, compared to reporting ID results alone (Huang TOF MS is also challenging, though some studies have shown that et al., 2013; Nagel et al., 2014; Perez et al., 2014; Lockwood et al., can be feasible (Mahe et al., 2014; Sakarikou et al., 2018) [P2 p3]
- Their results showed that MALDI-TOF MS rapidly and accurately classified Many studies have shown that MALDI-TOF MS equalled or unknown species of the genus *Aeromonas*, which was suitable for even surpassed the conventional diagnostic methods in speed environmental monitoring [P3 p5]
- et al., 2012; Marko et al., 2012; Martiny et al., 2012; Alby Microbial identification is achieved by comparing the observed et al., 2013; Karpanoja et al., 2014; Mather et al., 2014; spectrum with those stored in a database [P2 p2]
- A protocol for detection of carbapenemase production by However, as p019 polypeptide was absent in a subset of blaKPC-MALDI-TOF MS was developed in *Bacteroides fragilis* strains harboring plasmids, negative results should be complemented by harboring the *cfiA* gene encoding for the carbapenemase confirmatory tests [P1 p2]

Methods And Evidence Base

- to the detection of antibiotic-resistance (Figure 1) resulted from the observation that the hydrolysis of the -lactam ring Using the ClinProtTools analysis software (v3.0; Bruker after exposure of -lactam antibiotics to -lactamase producing Daltonics)

to investigate possible differences between the (aerobic and anaerobic) bacteria could be revealed in mass protein patterns of KPC-producing and non-KPC-producing spectra by a decrease of the peak corresponding to the antibiotic K [P1 p2]

- For *Candida albicans*, the minimal profile change concentrations (MPCCs) [i.e., the minimum Another MALDI-TOF MS-based method requires incubation drug concentration at which an alteration of mass spectra of bacteria with different concentrations of antibiotic, recovery can be detected, has been highly correlated with the MICs of microorganisms before MS analysis (Li et al., 2018) using the (Marinach et al., 2009)] [P1 p5]

- This article was submitted to Keywords: blood culture, MALDI-TOF mass spectrometry, bacterial identification, fungal identification, Systems Microbiology, antimicrobial resistance a section of the journal *Frontiers in Microbiology* INTRODUCTION Received: 06 December 2017 The application of mass spectrometry (MS) to bacterial identification was firstly proposed Accepted: 08 May 2018 by Anhalt and Fenselau (1975), based on the observation that different mass spectra were Published: 29 May 2018 o... [P2 p1]

- aureus for analysis make MALDI-TOF MS one of the most promising lineages using MALDI-TOF MS (Du et al., 2002; Schlebusch techniques for the identification of microbial pathogens directly et al., 2010; Gagnaire et al., 2012; Josten et al., 2013; Sparbier from positive BCs (La Scola and Raoult, 2009; Moussaoui et al., et al., 2013; Wang et al., 2013) [P2 p3]

- The presence of PSM-mec could be bacteria from BC using serum separator tubes (SSTs), which is detected by MALDI-TOF MS as a 2415 ? 2 m/z peak (Rhoads simple, rapid and yields a relatively high recovery of bacteria et al., 2016) [P2 p3]

- However, the time saving spectra, fungal cells may need to be lysed in 70% formic acid using this method over the reference AFST method (Clinical and and/or by mechanical disruption in a bead-beater (Amiri-Eliasi Laboratory Standards Institute, 2012) was modest (overnight vs [P2 p5]

Limitations And Gaps

- Lack of identification and misidentification species ID, score values 1.70 and <2.00 are reported as low were mainly due to insufficient database entries for some confidence ID, and scores <1.7 are considered as unreliable bacterial species but these limitations could be overcome by (no ID) [P2 p2]

- aeruginosa for resistance to meropenem, Strength and limitation of this method will be discussed below [P1 p5]

- The limitation of the technology is that identification of new isolates is Jugsharan S [P3 p1]

- MATERIAL GAP: Deep-read coverage: 3 of 25 screened records produced stored PDFs.

- MATERIAL GAP: PDF failures: 12 blocked or failed PDF attempts.

Literature Table

| Paper | Title | Year | Venue | Evidence Count |

| --- | --- | --- | --- | --- |

| P1 | Detection of Antibiotic-Resistance by MALDI-TOF Mass Spectrometry: An Expanding Area | 2020 | Frontiers in Cellular and Infection Microbiology | 25 |

| P2 | Recent Advances and Ongoing Challenges in the Diagnosis of Microbial Infections by MALDI-TOF Mass Spectrometry | 2018 | Frontiers in Microbiology | 25 |

| P3 | MALDI-TOF mass spectrometry: an emerging technology for microbial identification and diagnosis | 2015 | Frontiers in Microbiology | 25 |

Evidence Table

- Supported extracted claims: 75
- Parsed paper references: 3
- Screened records: 25
- Blocked records: 0
- Deep-read PDFs: 3

See evidence_table.csv for the full page-anchored evidence table.

Citation Audit

- Citation check: pass
- Supported paragraphs: 4
- Blocked paragraphs: 0
- Material gaps: 2